

Cedar Bluff Quarry Haul Road - Least-Cost Route Report (route CB-HR-01)

Basis: drawing CB-HR C-001/C-002 Rev B and the two rasters it registers. Horizontal CRS WGS 84 / UTM zone 14N (EPSG:32614); lengths are horizontal. Status: route study only, not for construction.

1. Result

Quantity	Value
Total accumulated route cost (USD)	2,339,944.02
Horizontal length, start cell centre to end cell centre (m)	4,921.44
Vertices (DEM cell centres) / moves	419 / 418 (179 diagonal)
Start cell centre (T1 on CR-114)	E 583155.00 N 3349105.00 z 374.54 m
End cell centre (G1 quarry gate)	E 586315.00 N 3350995.00 z 462.35 m
Creek crossing	Approved window X-2, culvert on 2 watercourse cells at N 3349605 (E 584605 to 584615)
Maximum grade on any move (limit 10.0 %)	9.48 % (uphill towards G1); steepest downhill -8.14 %
Northernmost point (route passes round the north end of Cedar Bluff Ridge)	E 585385 N 3351355
Closest approach to HS-1 recorded point (exclusion 150 m)	424 m

2. Method

Terrain. DEM DN converted to elevation with $\text{Elevation(ft)} = 1150.00 + 0.01 \times \text{DN}$ (Table 1), then to metres with the US survey foot (1 ft = 1200/3937 m). The drawing's vertical unit is feet while the grid is in metres; without the conversion every grade is 3.28 times too large and no route under 10 % exists.

Georeferencing. Both rasters are pixel-is-area with the upper-left corner of the upper-left pixel as registered on C-002 Table 1. DEM cell centres are at E 583 005 + 10i, N 3 351 495 - 10j. The land-cover raster has its own origin (E 582 900, N 3 351 620) and 20 m pixels; its edges coincide with DEM cell edges, so each land-cover pixel covers exactly 2 x 2 DEM cells, and each DEM cell takes the class of the land-cover pixel containing its centre. Colours were matched on exact RGB (every pixel matched one of the 8 classes).

Control points. Table 3 WGS 84 lat/long converted to EPSG:32614 with PROJ (pyproj). T1 -> E 583 153.23, N 3 349 102.62 (cell centre E 583 155, N 3 349 105); G1 -> E 586 318.42, N 3 350 996.91 (cell centre E 586 315, N 3 350 995).

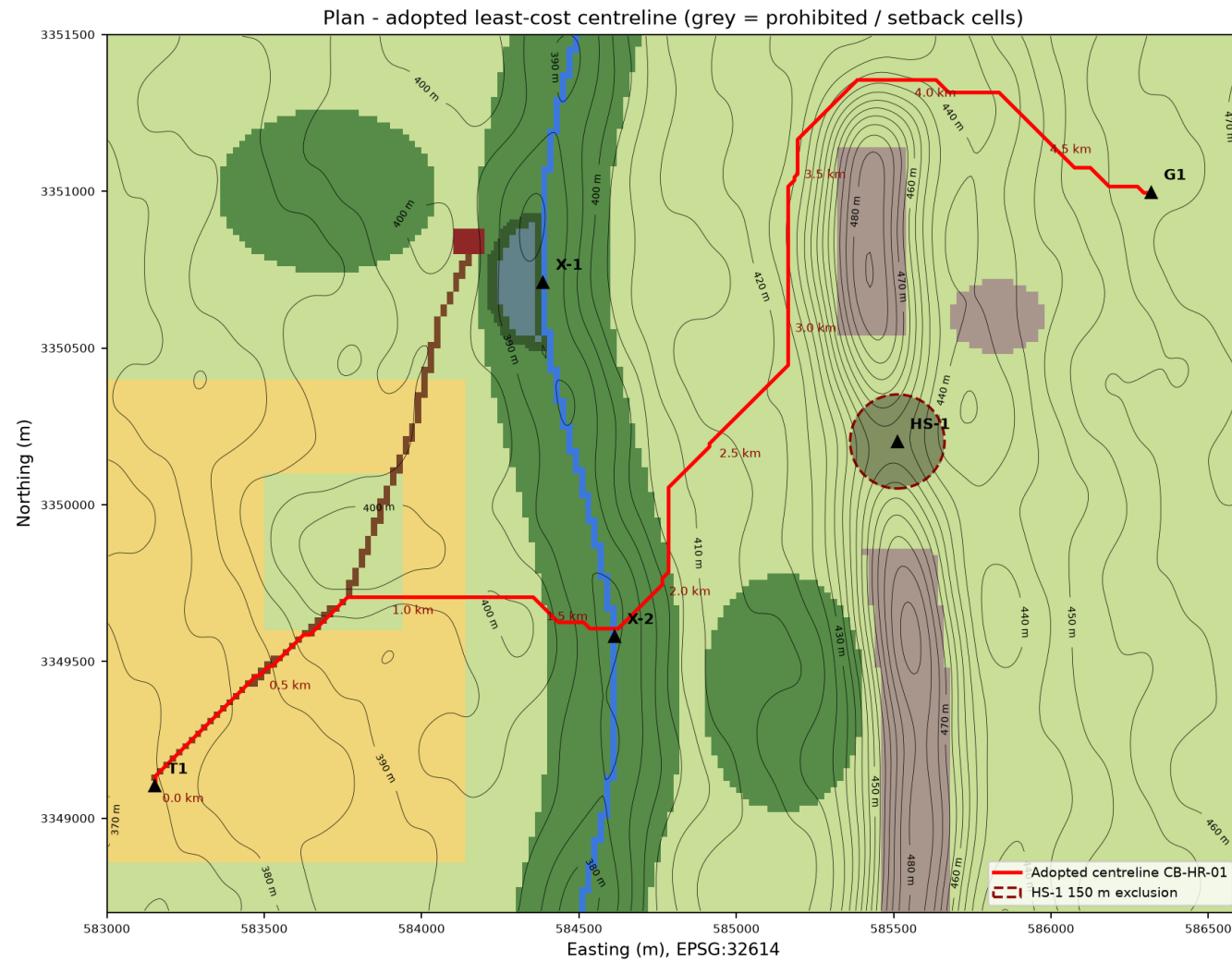
Constraints. Prohibited: wetland, farmstead, and watercourse cells except those whose centre is within 30.0 m of X-1 or X-2 (costed at 3 800 USD/m); every cell whose centre is within 30.0 m of the mapped wetland boundary (the edges of the wetland pixels); every cell whose centre is within 150.0 m of HS-1. Woodland at the Rev B rate of 940 USD/m.

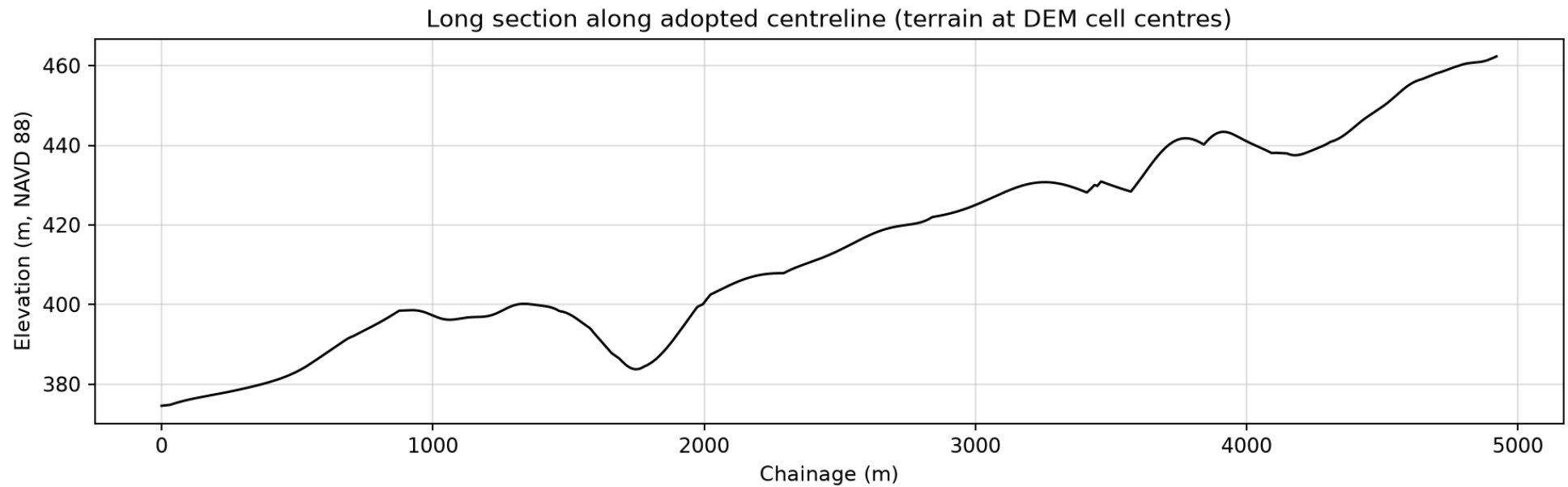
Search. Dijkstra accumulated-cost search on the 10 m DEM grid, 8-connected (queen's case). A move of horizontal length d (10 m or 14.142 m) between cells i and j costs $d \times (c_i + c_j)/2 \times f(g)$, where $g = |z_j - z_i| / d$ is the grade along that move and f is the brief's grade factor; any move with g above 10.0 % is not allowed. Grade therefore depends on the direction of travel: a hillside too steep to climb straight up can still be traversed on the diagonal. A single isotropic terrain-slope raster does not model this and gives a different route.

Rates and constraint values applied (C-002 Rev B)

Item applied	Value
Grassland / pasture	420 USD/m
Cultivated cropland	480 USD/m
Woodland (Rev B; Rev A 690 withdrawn)	940 USD/m
Rock outcrop	1,150 USD/m
Existing gravel track	160 USD/m
Watercourse cell within 30.0 m of X-1 / X-2 centre (culvert)	3,800 USD/m
Watercourse elsewhere, wetland, farmstead	prohibited
Wetland W-1 setback from mapped boundary	30.0 m, prohibited
HS-1 exclusion radius	150.0 m, prohibited
Maximum grade on any move	10.0 %
Grade factor f(g)	1.00 ($g \leq 3$); $1.00 + 0.05(g - 3)$ ($3 < g \leq 6$); $1.15 + 0.12(g - 6)$ ($6 < g \leq 10$)

3. Plan and long section





4. Breakdown

Land-cover class	Length (m)	Cost (USD)
Grassland / pasture	3,343.4	1,458,033
Cultivated cropland	215.0	103,298
Woodland	467.0	558,533
Existing gravel track	876.1	143,426
Watercourse at X-2 (culvert)	20.0	76,654
Total	4,921.4	2,339,944

Each move's length is split equally between its two end cells; its cost is split in proportion to the two cells' unit rates.

Grade band (abs)	0-3 %	3-6 %	6-8 %	8-10 %
Length (m)	2,863.1	1,570.7	223.1	264.6

5. Options and checks

Crossing X-1 is not feasible. The only land between wetland W-1 and Cedar Branch at X-1 is one 20 m land-cover column, and all of it lies within the 30 m wetland setback. X-1 cannot be reached from the west bank, and with X-2 removed the search finds no compliant route. The northern end of the farm track (towards the farmstead) and the 2025 indicative route through X-1 and HS-1 are therefore not viable. The adopted route uses the southern 0.88 km of the track and then leaves it towards X-2. **Cedar Bluff Ridge saddle:** the lowest pass over the ridge lies inside the HS-1 150 m exclusion. Without the exclusion the least-cost route would take the saddle (run below). With it, the adopted route climbs round the north end of the ridge. The runs below were made only as sensitivity checks; each breaks a stated constraint and none is an option.

Run	Cost (USD)	Length (m)	Crossing
Adopted route (all constraints)	2,339,944	4,921.4	X-2
Check: without 30 m wetland setback	1,989,674	4,664.3	X-1
Check: without HS-1 exclusion	2,105,916	4,260.9	X-2
Check: Rev A woodland rate 690 USD/m	2,191,398	4,921.4	X-2

6. Compliance

No vertex lies in a prohibited, setback or HS-1 cell; the only watercourse cells used are within 30 m of X-2. Every move is at or below 10.0 % (maximum 9.48 %).

Limitations: the centreline is a 10 m raster least-cost path, i.e. a corridor for alignment design, not a geometric design. Curvature, sight distance, earthworks balance and culvert hydraulics are not assessed. In flat, uniform-cost ground several paths tie on cost; tie-break and foot-definition checks gave the same cost and length, with paths no more than 10 m from this one.

Deliverables: CB_HaulRoad_Centreline.gpkg (layer 'centreline', one LineString, EPSG:32614), CB_HaulRoad_Vertices.csv (one row per vertex, start to end), this report.